

Methodology

1. Data Preprocessing

- The dataset was imported and inspected using `glimpse()`, `skim()`, and summary statistics. Variable types were corrected by converting `status_label` to a factor and ensuring all financial variables were numeric. No missing values were detected across the 21 variables. Outlier inspection using boxplots confirmed the presence of extreme financial values, which were retained to preserve real-world variation. A new engineered feature—the *financial stress index*—was created using the formula:

$$\text{Financial Stress Index} = \frac{\text{Total Liabilities}}{\text{Total Assets}} - \frac{\text{Net Income}}{\text{Total Assets}}$$

This variable captures the tension between leverage and profitability, two core drivers of bankruptcy risk.

2. Exploratory Data Analysis

- Univariate EDA included frequency tables, histograms, and a pie chart showing the distribution of bankruptcy outcomes. Multivariate EDA used boxplots and a correlation heatmap to examine relationships among financial variables. Strong positive correlations were observed among profitability and cash-flow metrics, while leverage-related variables showed distinct patterns associated with distress. These insights informed feature selection and interpretation of clustering results.

3. Data Transformation

- All numeric variables were standardized using z-score scaling to ensure equal weighting in Euclidean distance calculations. The dataset was split into training and testing subsets to allow post-clustering evaluation against true bankruptcy labels without leakage.

4. Clustering Model

- K-means clustering was applied with $k = 2$ to reflect the binary nature of bankruptcy outcomes. The model was trained on the scaled numeric variables, including the engineered financial stress index. Cluster assignments were compared to actual bankruptcy labels to determine which cluster represented distressed firms. Cluster 2 contained nearly all failed firms (5,220) and the majority of solvent firms, while Cluster 1 represented a small subset of financially strong firms.

5. Model Evaluation

- Cluster assignments were mapped to bankruptcy labels to compute performance metrics. Accuracy, precision, recall, and F1-score were calculated using the test set. The model achieved:
 - **Accuracy:** 93.8%
 - **Precision:** 93.8%
 - **Recall:** 99.9%
 - **F1-score:** 0.968

The extremely high recall indicates that the clustering model successfully identified nearly all failed firms, demonstrating strong alignment between financial structure and bankruptcy outcomes.